Exploring DeepFakes: Face Reenactment Report

October 12, 2026

1 Diffusion-Based Head Swap

1.1 Scope and hardware

We wanted to swap a video face with one from a selfie. The run had to stay on one laptop with a 4 GB NVIDIA GPU and 8 GB of system RAM. We sent no data to a cloud service.

The output had to keep the source motion, expression, lighting and background. Only the head should change. The clip is 96 frames at 512x512 and lasts 3.2 seconds.

1.2 How we measured

Every number comes from one script run on the same 96 frames, and the results table reports it. Face change is how much the face region moves from one frame to the next. The unswapped footage sets the level to match, so a value far below it means the smoother removed real motion too. Consecutive identity is how similar a face is to the one before it, measured in a face recognition embedding, so higher is more stable. Identity to the portrait measures similarity to the reference portrait; the unswapped footage scores near zero because it is a different person. Hair noise is how much the hair region flickers, and lower is better; p95 marks the worst frames, the value 95 percent stay under.

1.3 Pipeline overview

The flow has four stages. Generation transfers the identity, restoration cleans each face with GFPGAN, smoothing removes the boil, and reassembly attaches the audio. Only the last stage writes a video file.

The boil is the flicker that comes from generating each frame on its own. Figure 1 shows the chain in order, with the cost of each stage.

1.4 How the diffusion model works

A diffusion model learns one job: look at a noisy image and say what noise is in it. Training adds noise to clean images in small steps until only noise is left. The network guesses the noise at each level, and the guess is compared with the truth. The weights are then nudged.

After training, the same network runs backwards. It starts from pure noise and removes its own guess at every step, so an image forms. The forward direction only exists for training, and we never run it to generate. Figure 2 draws both directions.

We did not train the base model; we used a released checkpoint. The released model is step distilled, so four steps are enough for it. We could not tell the difference between four steps and eight, and eight took twice the time.

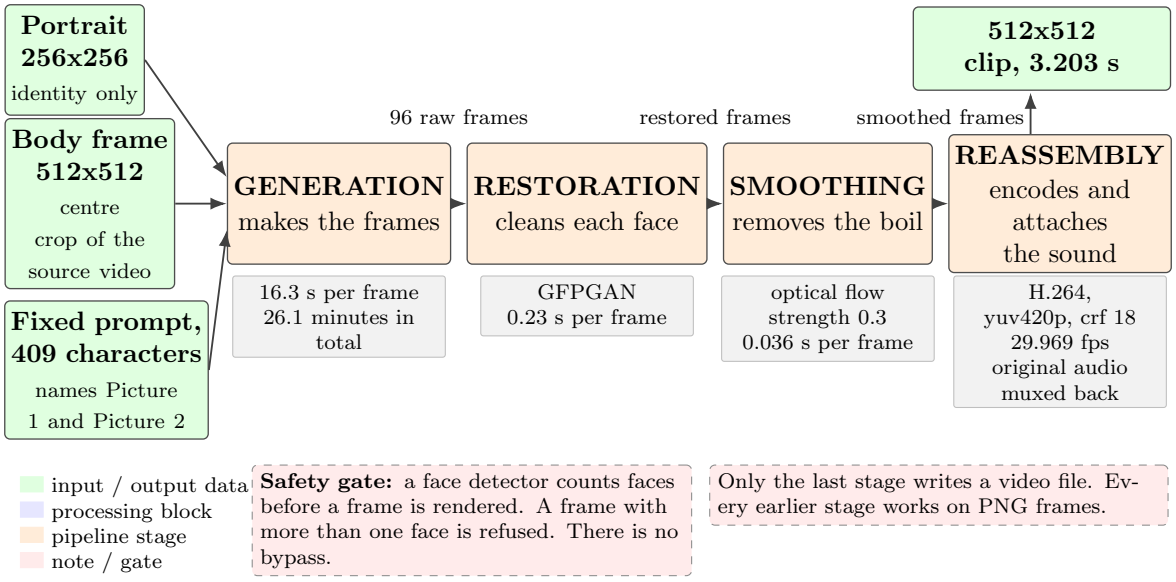


Figure 1: The four-stage pipeline: generation, restoration, smoothing and reassembly, with the cost measured for each stage.

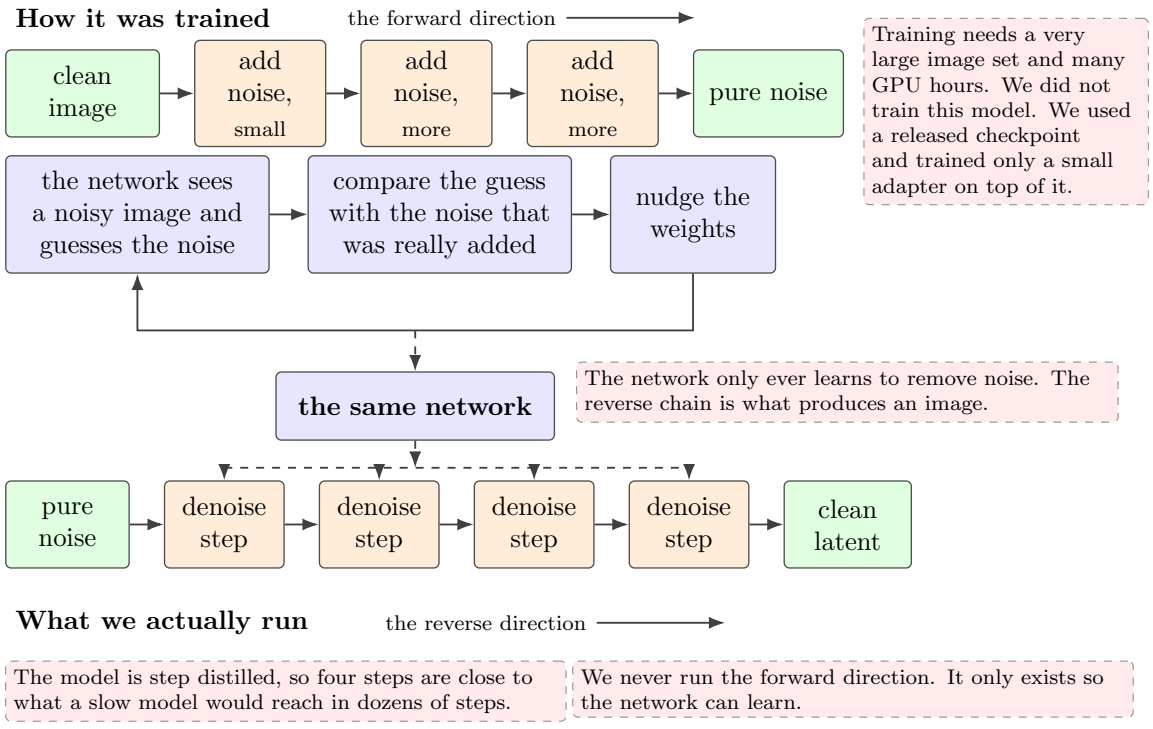


Figure 2: The two directions of the diffusion model. The forward direction adds noise and is only used in training; the reverse direction removes noise and produces the image. Both directions use one and the same network.

1.5 The denoiser and the LoRA adapter

The denoiser is a stack of transformer blocks (Figure 3). A latent token of 128 channels is projected to width 3072. It passes through five double-stream blocks, then 20 single-stream blocks and a final layer. In the first part the text and image tokens stay apart, and they meet through joint attention.

Later the two streams are joined into one. The final layer projects back to 128 channels for

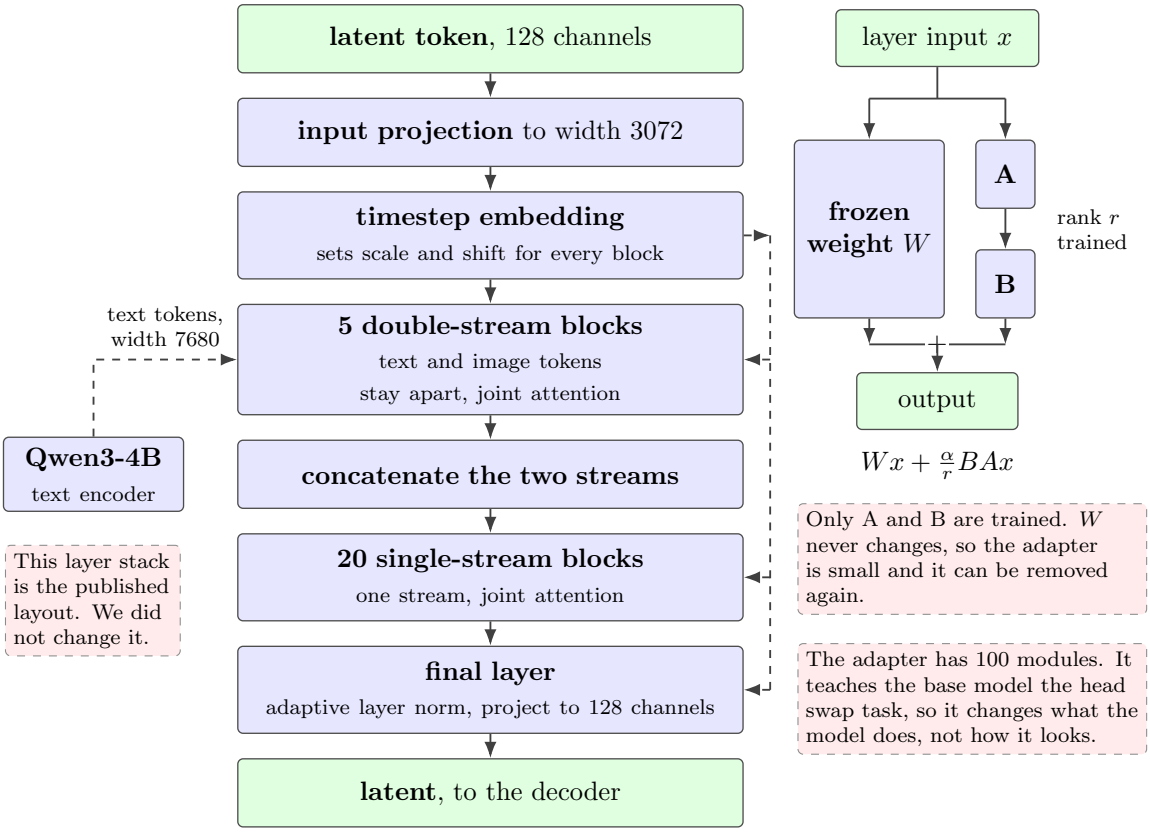


Figure 3: The denoiser as a stack of layers. The text encoder and the timestep embedding enter as conditioning, and the adapter adds a small trained branch beside a frozen weight.

the decoder.

The adapter is a small pair of matrices, A and B, trained beside each frozen weight. The block output adds a scaled product of A and B to the frozen term. Only A and B are trained, so the base weights never change. The adapter carries the head swap behaviour.

1.6 Generation

Generation (Figure 4) covers conditioning, denoising and decoding. A fixed prompt of 409 characters names Picture 1 as the body and Picture 2 as the head.

Qwen3-4B encodes the prompt once per session on the CPU, about 80 seconds, and its memory is then freed. Both pictures become VAE latents: the 512x512 body crop a 64x64x32 grid of about 1024 tokens, the 256x256 portrait about 256 tokens.

FLUX.2-klein-4B renders each frame in four denoising steps (Figure 5). The text embeddings and the two image latents guide every step. A community-trained LoRA adapter of 100 modules carries the head swap.

The same VAE decodes the final latent into pixels. Picture order is part of the model contract, so a reversed order gives the wrong head.

The generator runs as a 4-bit GGUF file, and 256 pixels is the largest reference that fits. A detector refuses a frame with more than one face, and the gate has no bypass.

The output is 96 frames from 5.8 to 9.0 seconds of the source, all with one fixed seed. Each frame took 16.3 seconds, so the stage used 26.1 minutes.

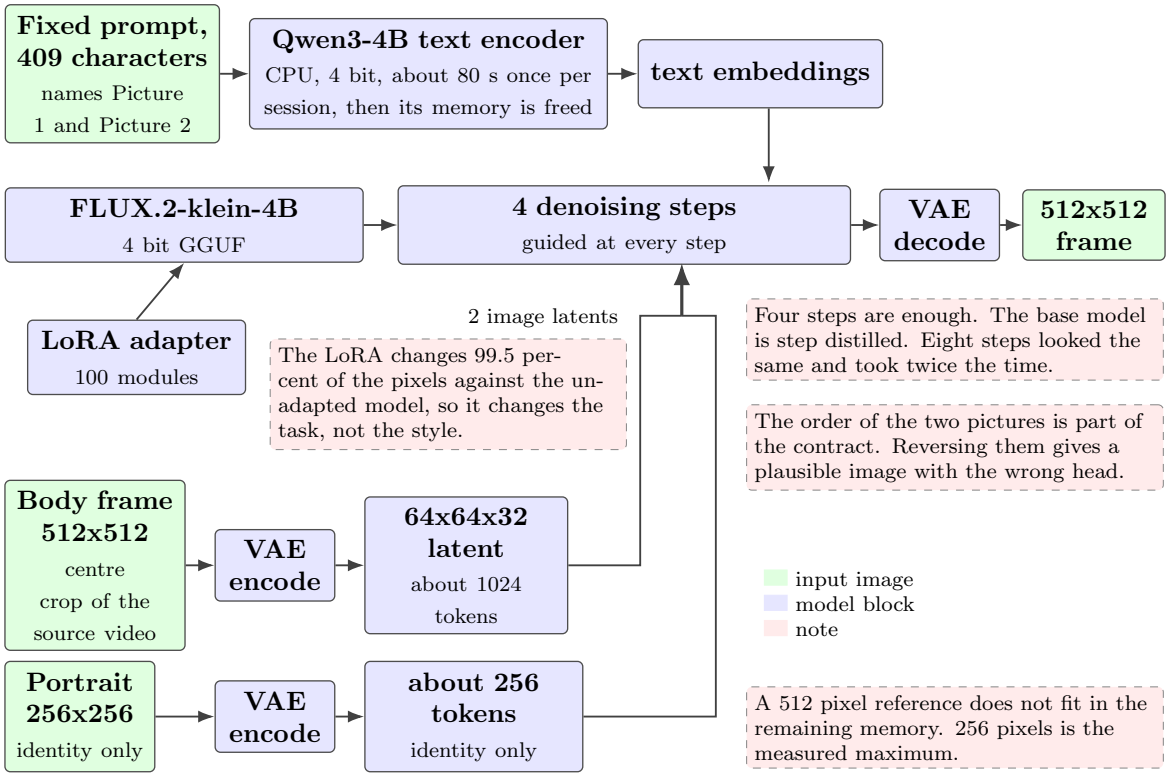


Figure 4: Inside the generation stage: text conditioning, the image latents, the denoiser with its LoRA adapter, and the four denoising steps.

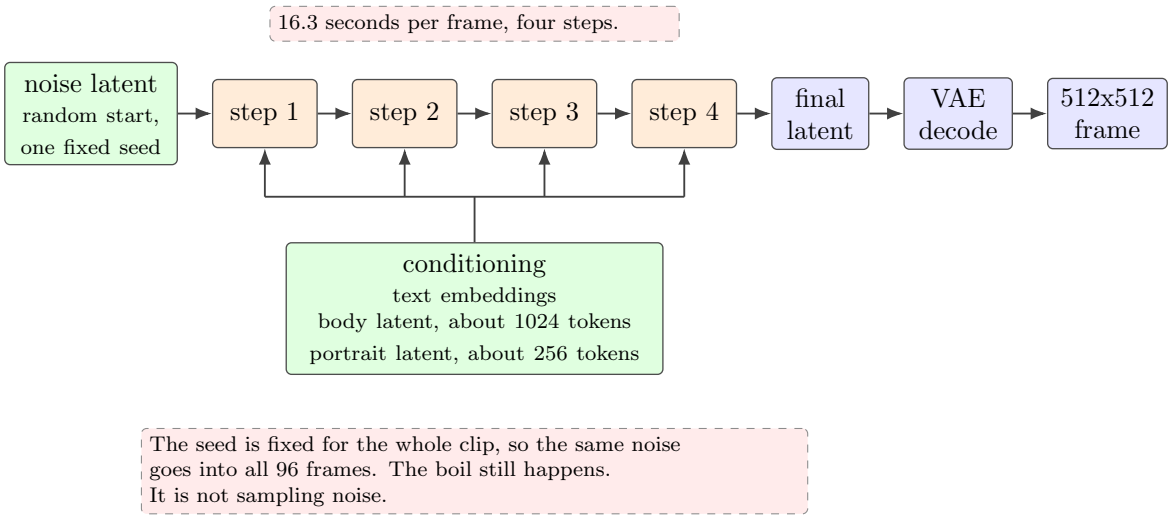


Figure 5: One generation: the denoising loop. The model starts from a noise latent, runs four denoising steps that receive the conditioning at every step, and decodes the final latent into a 512x512 frame.

1.6.1 Adapter version comparison

The recorded clip uses adapter v1 throughout. A newer adapter, v1.1, was released during the work, and we tested two prompt variants, A and B. Table 1 compares the two versions.

v1.1 ran, but it went far past the memory of the card, which is why it slowed down so much. The extra time was not worth it for us.

We judged the look by eye and preferred v1, so the clip kept the v1 output. No metric script

	v1 (recorded)	v1.1 (new)
Seconds per frame	16.34	about 199, twelve times slower
Peak VRAM allocated	3.60 GiB	4.68 GiB, past the 4.0 GiB of the card
Peak system RAM	5.48 GiB	5.50 GiB
Cost for 96 frames	26 minutes	about 5.3 hours

Table 1: Adapter v1 against v1.1 on the same machine; only v1 was measured with the metric script.

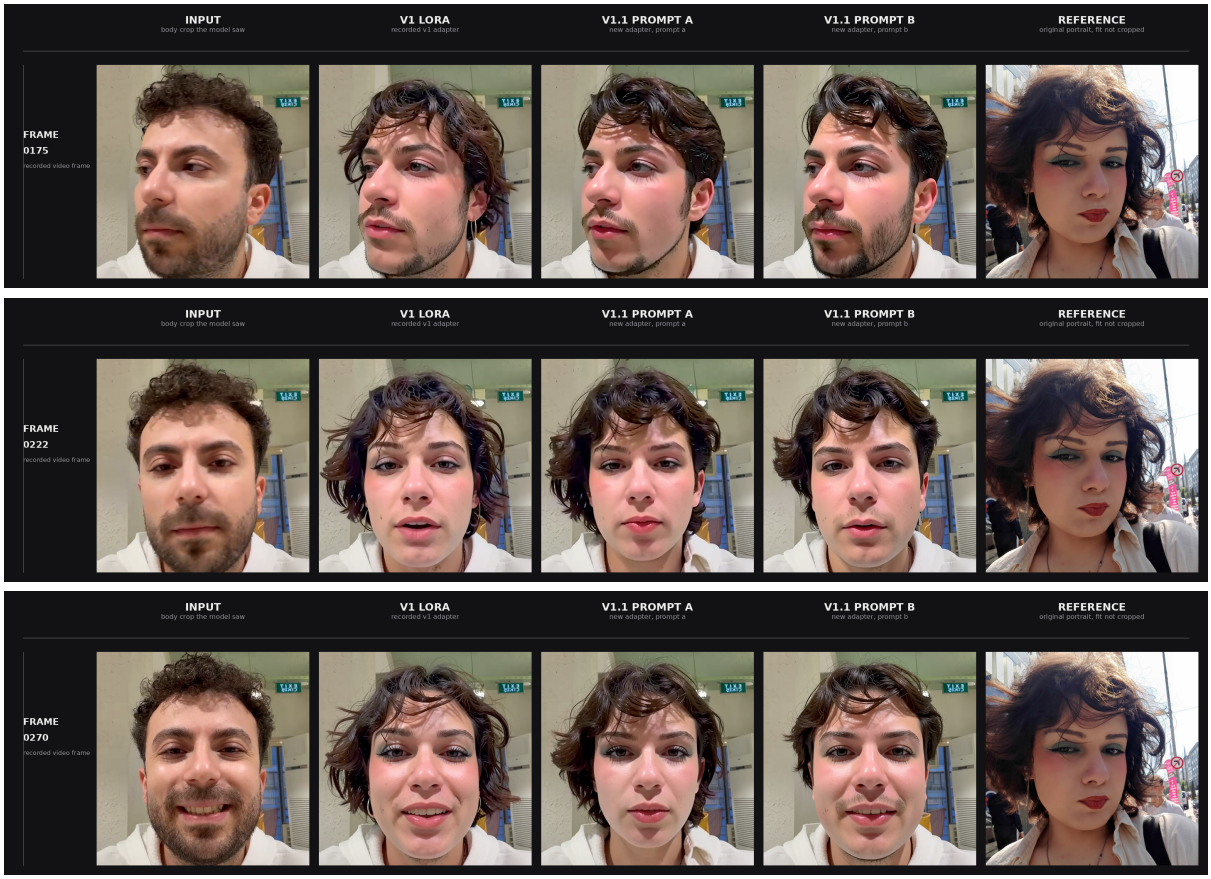


Figure 6: Each row is one source frame, the frame number printed inside. The five columns run from the input body crop through the v1 adapter, v1.1 with prompt A, v1.1 with prompt B, to the reference portrait; only the adapter version and the prompt variant change.

was run on v1.1, so every other number here comes from v1. The grids in Figure 6 are the only evidence.

1.7 Face restoration

GFPGAN v1.4 cleans every frame (Figure 7). It detects and aligns the face, runs a restoration network, and composites the crop back. The weights are official, but we built the network in project code because the published wrappers do not import.

The stage costs 0.23 seconds per frame. The frames looked cleaner to us, and the mean face change came down, though the tail improved less.

Hair noise fell slightly, a small gain on its own. The fidelity weight does nothing, because the official inference path passes no weight argument. Restoration cannot remove flicker, because it sees one frame at a time.

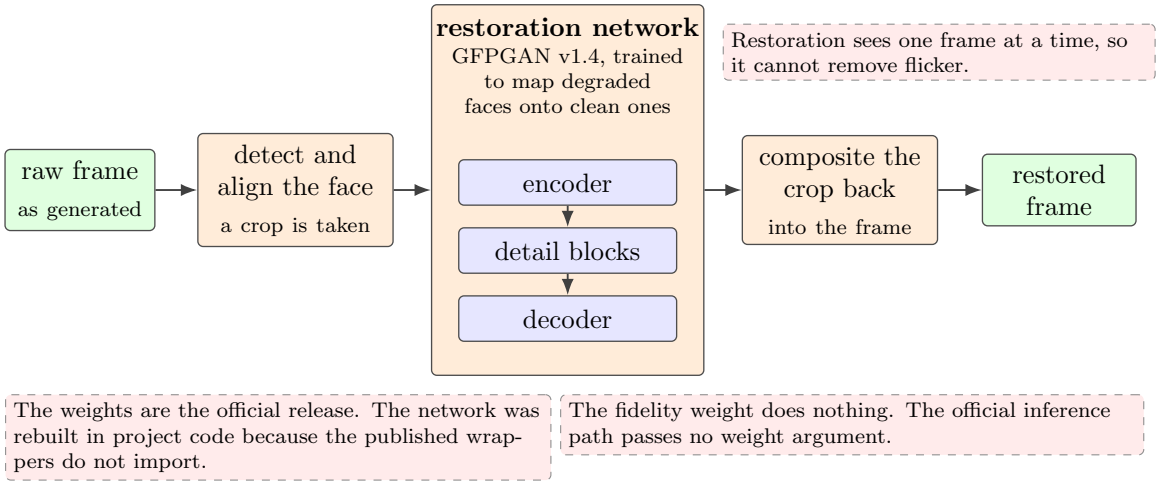


Figure 7: The face restoration stage. Each raw frame is taken on its own: the face is detected and aligned, a crop runs through the restoration network, and the restored crop is composited back into the frame.

Frames smoothed	Strength	Face change	Distance from the control	p95	Consecutive identity	Hair noise
Raw	0.3	6.9599	+0.3034	16.2944	0.9330	32.2358
Raw	0.6	6.6663	+0.0098	15.6581	0.9391	32.0603
Restored, then flow	0.3	6.6303	-0.0262	15.2802	0.9377	31.7306
Restored, then flow	0.6	6.3071	-0.3494	14.7575	0.9431	31.5448

Table 2: The smoothing sweep: two frame sets at two strengths, all measured on the same 96 frames.

1.8 Temporal smoothing

The smoothing stage compares frames with each other and removes the boil (Figure 8). The per-pixel deflicker averages each pixel with its temporal neighbours, and it is too strong. It costs 0.007 seconds per frame.

The deflicker pushes the face change below the unswapped level, so it removes real motion as well as the boil. Optical flow estimates the dense motion between neighbouring frames and warps them onto the current one. Because it follows the same surface point across time, a moving head does not smear. It costs 0.036 seconds per frame.

The control fixes the real motion level at 6.66, and the distance column is the face change minus that level. At strength 0.3 the chained pass comes close enough to the control, and the raw pass stays above it. A stronger setting sits below the control, which is over-smoothing.

The recommended output is the chained set at strength 0.3. It beats the flow-only variant on face change, on the tail and on hair noise. Only consecutive identity is slightly lower. The deflicker reaches its low face change only by destroying motion, so we kept flow.

1.9 Reassembly

The encoder turns a frame set into a playable clip (Figure 9). It writes H.264 at 29.969 fps, muxes the original audio back, and trims both streams. Its output is a 512x512 clip of 3.203 seconds.

The encode adds nothing to the pixels. In the raw swapped set the face boils, and the results table holds the values.

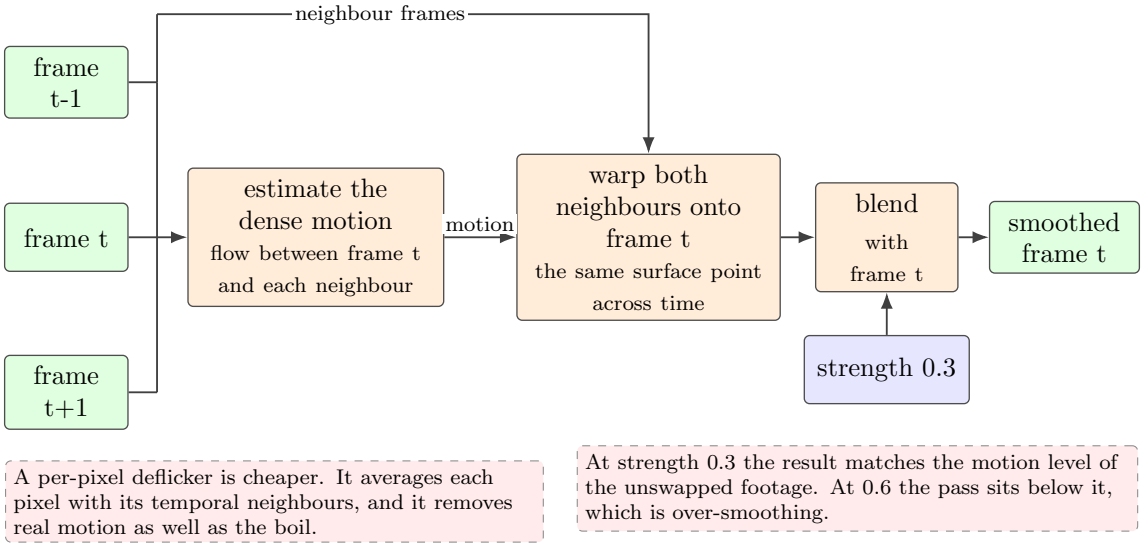


Figure 8: The temporal smoothing stage. Each frame is compared with its two neighbours: the dense motion is estimated, the neighbours are warped onto frame t and the three are blended at strength 0.3.

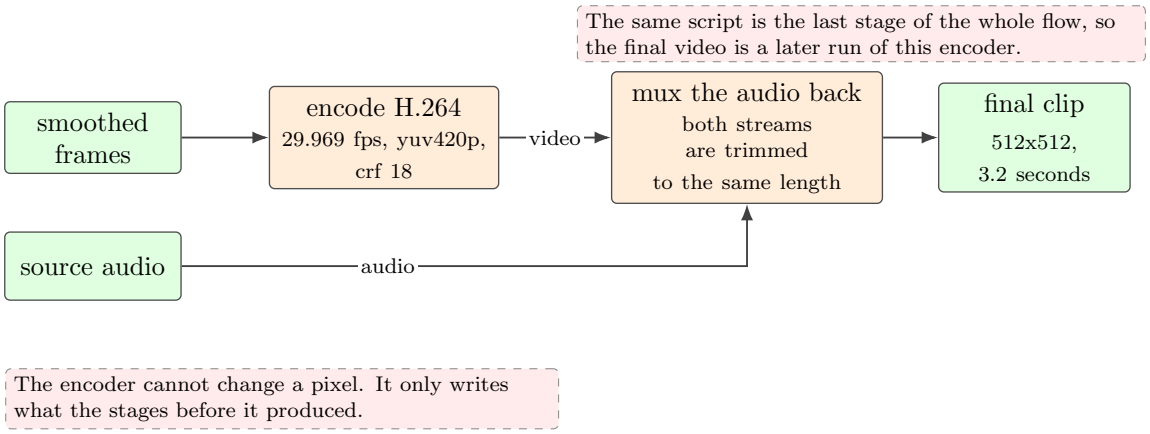


Figure 9: The reassembly stage. The smoothed frames are encoded as H.264 at the source rate and the original audio is muxed back; both streams are trimmed to the same length.

1.10 Results

All numbers come from one script on the same 96 frames. Lower hair noise is better, and higher identity values are better. Face change should sit near the control level, because a value below it means the smoother removed real motion. The control row is the unswapped source footage.

Identity to the portrait confirms the swap direction. The unswapped footage scores near zero, and every swapped variant scores positive. The recommended set held up well against the control on the mean values. Figure 10 shows the effect of each stage on one frame.

1.11 Problems and decisions

Attaching the adapter before the first forward pass cost 46 to 48 seconds per frame. Attaching it after one plain pass cost 16 seconds per frame. Peak VRAM was the same in both orders.

We standardised on the fast order. The internal cause was not established, but the effect reproduces exactly and explains an earlier timing discrepancy.

One fixed seed covers the whole clip, so the noise input is constant. The boil still happens,

Stage	Face change (mean)	Face change (p95)	Consecutive identity	Identity to the portrait	Hair noise
Control (unswapped source footage)	6.6565	12.3488	0.9852	−0.0356	31.5999
Raw head swap	7.4510	17.3794	0.9228	0.2153	32.5703
Restoration	7.1602	16.2097	0.9273	0.2153	32.0721
Flow on the raw frames	6.6663	15.6581	0.9391	0.2148	32.0603
Restoration then flow, strength 0.3 (recommended)	6.6303	15.2802	0.9377	0.2152	31.7306
Deflicker (alternative)	5.6269	12.6650	0.9357	0.2106	31.8960

Table 3: All measured variants on the same 96 frames.



body input raw swap after restoration after smoothing

Figure 10: The same frame 0175 at four points of the flow; the four images are all 512x512.

because the model is sensitive to small input changes. Only a pass across frames reduces it.

Memory set the last limit; larger references do not fit. The largest reference we could use is 256 pixels.

1.12 Limits

The results table shows a mean at the control level, but the tail stays high, so the picture still jumped on some frames. Each step works inside a frame or a short window, so no step removed that tail. The 256-pixel reference maximum is the second limit on this card.